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The table-top DiaDNA machine

Blind trust has been at the core of gem and jewellery purchases in India. But what happens when that trust is shaken? You send a ring to be re-sized and a few years later discover that the natural diamond was swapped for a synthetic or a lower grade stone. Within the industry, pilferage in secure diamond cutting facilities is a legitimate concern. Reports of cutters swapping a G-colour diamond (highest grade being D) for an H-colour and profiting from the margin is not uncommon.

What's the solution? The first thought is to get a diamond certificate or continuously keep grading the stone as it changes hands. But a paper report can be forged, unique numbers on diamond girdles, faked. Graders, bribed. Could there be a fool-proof method?

An Indian company (DiaDNA) has developed an AI-powered paternity test for diamonds, which is currently being used by the industry to prevent switching of stones at cutting and polishing facilities globally.

How does it work? The technology sits at the manufacturing stage of a diamond. A rough diamond is first scanned at an atomic level to map its DNA. The scan is not dependant on any inclusions or optical parameters which may be removed during the making of a diamond. You can cut/polish the rough diamond anywhere in the world. When you get the stone, you can place it back in the machine, to verify the origin of the cut diamond from the rough it was cut from and get the answer within 18-20 seconds.

"A diamond certificate tells you the weight, colour, and clarity," explains Mayank Jain, CEO of DiaDNA. "What it cannot tell you is whether the stone in front of you is the same stone that certificate was made out to. Every diamond of that grade looks identical to a buyer, and the paper travels separately from the stone. Our paternity test closes that gap. The evidence sits inside the diamond, not in a document that can be swapped, reissued or lost."

But the paternity test is not only for matching a rough diamond with one or more cut stones. The parent scan can also be done on a cut diamond. Many jewellers in India offer a 'buyback policy', allowing their clients to return a piece of jewellery they purchased and get an assured value back against a mutually agreed formula. But how would a jeweller know if a client is bringing back a ring (for example) with the same diamond that the jeweller sold? It could be a stone that bears the same dimensions but may be a synthetic or a marginally lower grade. And if the jeweller can tell the difference, how can they prove it?

"Today the technology sits at the manufacturing end," explains Jain. "The rough and the polished stone are scanned where they are cut, and the match is established there. The consumer doesn't operate anything; what reaches them is a record that travels with the stone. The consumer-facing leg (walking into a jewellery store and having your original stone re-verified), is what we're building next, table-top machines sitting at retail counters rather than only in the factories."

While the retail solution is still in development, several leading diamond firms have already adopted the technology to secure their pipeline, with many other pilots underway. Although DiaDNA's tech currently doesn't provide mine to market traceability, it is a revolutionary solution to close a significant gap that currently exists in the diamond sector.



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